Five-Stage Fourth-Order SDIRK

Skvortsov S54b non-geometric reference method

1 Butcher tableau

For $\dot{z} = f(t, z)$, the S54b tableau is

$$\begin{array}{c|ccccc} \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{8} & \frac{1}{8} & \frac{1}{4} & 0 & 0 \\ 1 & -\frac{3}{2} & \frac{3}{4} & \frac{3}{2} & \frac{1}{4} & 0 \\ 1 & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \\ \hline & 0 & \frac{1}{6} & \frac{2}{3} & -\frac{1}{12} & \frac{1}{4} \end{array}.$$

All five diagonal entries equal $\gamma = 1/4$, so the method is singly diagonally implicit. The final row of A equals b^T, making the method stiffly accurate.

The stages are solved sequentially:

$$Z_i = z_n + h \sum_{j<i} a_{ij} f(t_n + c_j h, Z_j) + h\gamma f(t_n + c_i h, Z_i), \tag{1}$$

$$z_{n+1} = z_n + h \sum_{i=1}^{5} b_i f(t_n + c_i h, Z_i). \tag{2}$$

Each stage therefore requires a nonlinear solve in the physical state dimension, rather than one simultaneous system containing all stages.

2 Fourth-order conditions

Writing $C = \mathrm{diag}(c)$ and using componentwise powers, the coefficients satisfy

$$\begin{array}{llll} b^T\mathbf{1} = 1, & b^T c = \frac{1}{2}, & b^T c^2 = \frac{1}{3}, & b^T Ac = \frac{1}{6}, \\ b^T c^3 = \frac{1}{4}, & b^T CAc = \frac{1}{8}, & b^T Ac^2 = \frac{1}{12}, & b^T A^2 c = \frac{1}{24}. \end{array}$$

These are the Runge–Kutta conditions through order four. For sufficiently smooth problems the local truncation error is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$.

3 Deliberately non-geometric structure

A Runge–Kutta method is symplectic for general canonical Hamiltonian systems only if

$$b_i a_{ij} + b_j a_{ji} - b_i b_j = 0 \qquad \text{for every } i, j.$$

The S54b coefficients do not satisfy this identity. For example, with $i = j = 5$ the left-hand side is $1/16$, not zero. Hence S54b is not a symplectic Runge–Kutta method.

Symmetry would require equality with the adjoint tableau, including

$$c_i = 1 - c_{6-i}, \qquad b_i = b_{6-i}, \qquad a_{ij} + a_{6-i,6-j} = b_{6-j}.$$

Already $c_1 = 1/4 \neq 1 - c_5 = 0$, so S54b is not symmetric. These failures make the method a useful same-order control in long-time comparisons against geometric fourth-order methods.

4 Newton implementation

For a known right-hand side r_i, one stage solves

$$R_i(Z_i) = Z_i - r_i - h\gamma f(t_n + c_i h, Z_i) = 0.$$

Newton corrections satisfy

$$\left[I - h\gamma D_z f(t_n + c_i h, Z_i^{(k)})\right] \delta_i^{(k)} = -R_i(Z_i^{(k)}).$$

Execution supplies f and its optional dense or particle-block Jacobian; guiding-centre systems supply independent 2×2 blocks. Configure differentiation with `solver=NewtonSolver( jacobian_method=...)`. Analytic mode uses the supplied derivative, finite-difference mode approximates it from f, and auto uses the supplied Jacobian when available. The solver also owns tolerances and iteration limits. Per-stage and per-step iterations, residual evaluations, residual norms, and effective tolerances are returned as diagnostics.

5 Reference

The coefficient table is the S54b method presented by L. M. Skvortsov, "Diagonally implicit Runge–Kutta methods for stiff problems," *Computational Mathematics and Mathematical Physics*, 46(12), 2110–2123 (2006), DOI: 10.1134/S0965542506120098.

6 Physical accepted states and passive energy tracking

For N particles with d physical components, every accepted snapshot contains $Z \in \mathbb{R}^{dN}$ and one scalar time t. An optional energy observer owns separate physical momentum $\kappa \in \mathbb{R}^N$. Thus GC accepted vectors have $2N$ components and FC vectors have $4N$, whether energy is tracked or not. A doubled method prepares temporary $(u, v) \in \mathbb{R}^{2dN}$ copies, allows them to separate, and projects to one physical result before acceptance. The representation stores only the physical size and never owns a clock, momentum, physical coordinate declaration, dynamics or solver.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, Z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, Z(t)) + \kappa(t) - H(t_0, Z_0).$$

GC split-map contributions are summed in temporary work and divided by two to obtain physical momentum. Tracking leaves physical stages, nonlinear stopping scales and adaptive acceptance unchanged. An autonomous Hamiltonian supplies an explicit zero momentum derivative.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired. Dynamics declare component order and position/velocity roles through immutable structural data; initial conditions and results retain those data independently. Execution injects $f(t, Z)$ and optional physical derivatives into a fresh method run. Methods own physical stages and projection equations, without retaining the dynamics object. Nonlinear tolerances, iteration limits, differentiation and iteration callbacks belong to the configured algebraic solver. Generic doubled copy operations live in `src/formulations/state.py`; classical, HBVM and adaptive methods need no state formulation.

Methods optionally return retained stage quadrature or a dense interpolant. The energy observer integrates momentum independently, using Hamiltonian functions supplied by execution. Controllers keep only physical accepted states; compiled JAX loops carry the observer accumulator separately. Observers receive physical states. Energy histories in `States.diagnostics` retain their existing labels; `extended_time` has shape (N, n_{samples}) and is broadcast from scalar sample times. Fixed shadow samples leave accepted states and counters unchanged. Adaptive quadrature uses accepted dense output and physical-only error control.

The five converged physical stages supply the passive momentum quadrature with the tableau weights.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Newton systems use the supplied particle-block structure. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.